4. Career Development (28)

AP CSA 11.0160001-63 - Niki Lauda Assignment:

Nikki Lauda became one of the most successful Formula 1 drivers in history not simply because he could drive a race car, but because he understood the factors that affected performance. Rather than focusing only on driving, he asked questions about weight, materials, reliability, and design.

In computer science, programming languages such as Java and Python, as well as AI tools, are just that: **tools**. The most successful computer scientists and software engineers first understand the problem, identify important variables and constraints, and then use technology to build solutions.

This activity asks you to think critically about the relationship between problem-solving and programming.

Video: **RUSH (2013) | Niki Lauda joins to BRM team (5min)**

Niki_Lauda_Exit_Ticket_APCSA.docxDownload Niki_Lauda_Exit_Ticket_APCSA.docx

Instructions

1. Read each question carefully.
2. Answer all questions completely and thoughtfully.
3. Focus on explaining your thinking, not on writing code.
4. Use examples from real-world situations when appropriate.
5. Submit your completed document

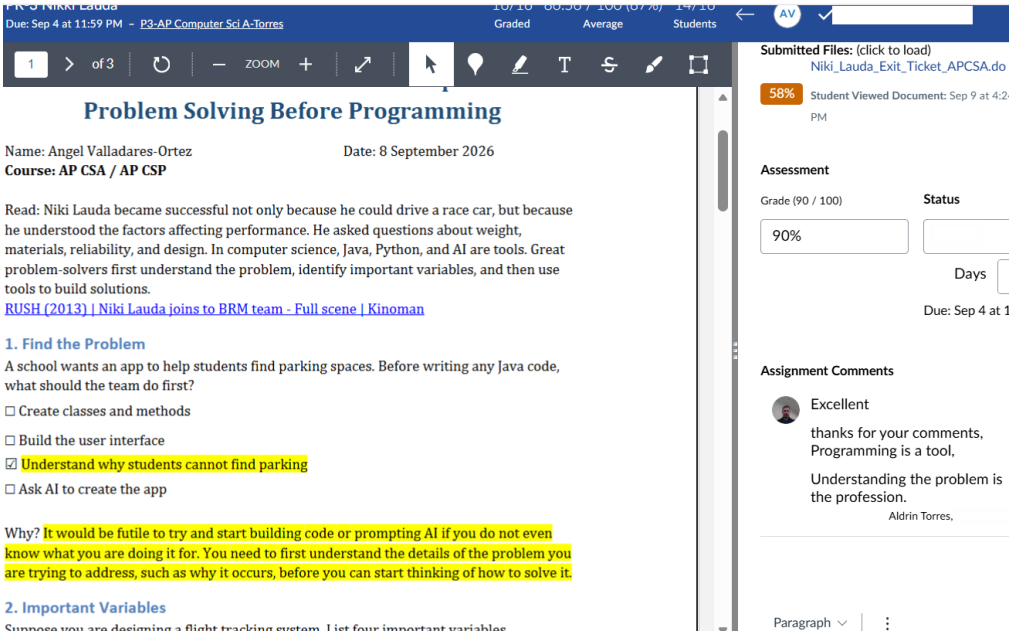
AI Expectations

You may use AI to clarify vocabulary or concepts if needed.

However, your responses must reflect **your own thinking and reasoning**.

Remember: AI can generate code, but it cannot automatically determine the most important problem to solve, or the needs of the user.

Your goal is to demonstrate that you understand the problem-solving process, not simply that you can generate an answer.



GD: Animation & Sim 11.4290001-61 - Gaming Industry Research:

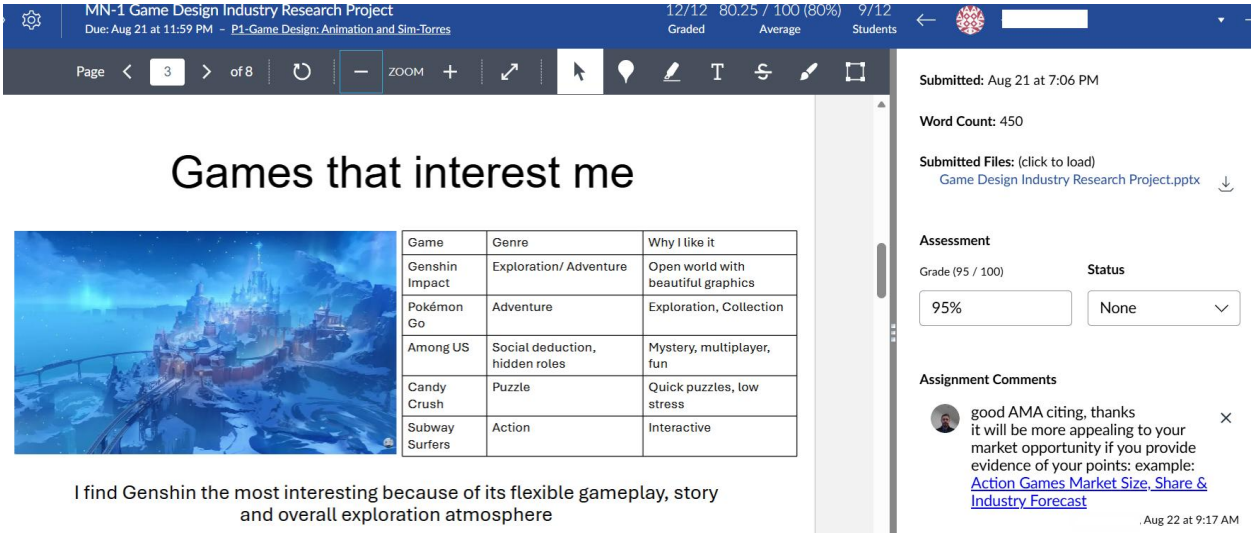
Maximum: 8 Slides Individual Assignment

Driving Question

If you were going to build a game, what type of game would you create and why would it have the best chance of success?

Assignment Goal

Research the game industry, explore different game categories, and identify a game type that interests you. Use industry information to justify why your selected game genre has market potential, career opportunities, and manageable ethical or legal concerns.



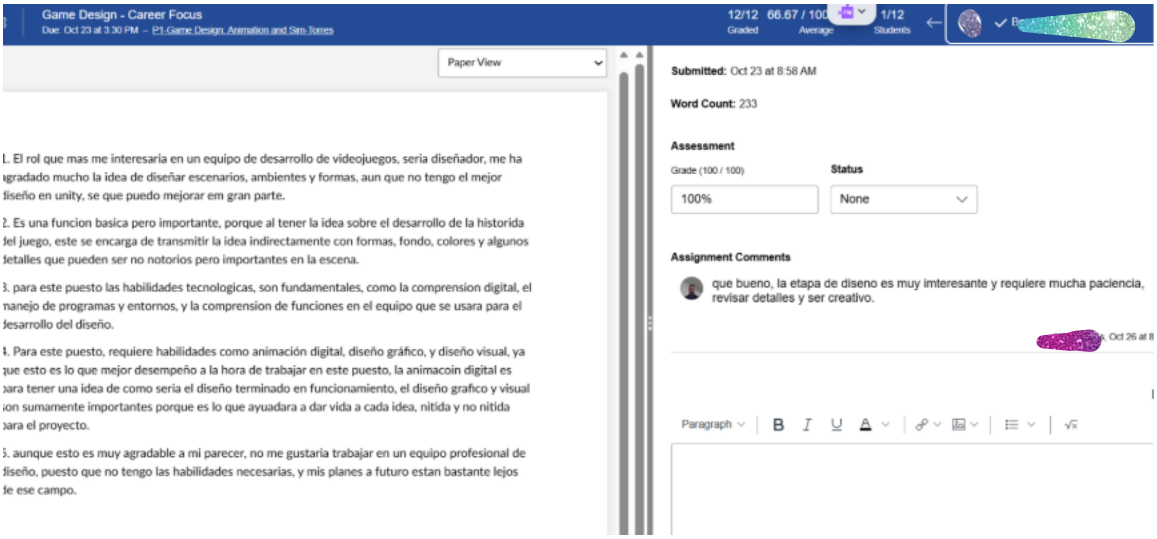
4. Career Development (28) latest

Game Design & Simulation

Activity Summary: Careers in the Video Game Industry activity. Students explored roles including game designer, programmer, artist, producer, QA tester, and audio designer while identifying technical skills, workplace responsibilities, and career pathways in the game industry.

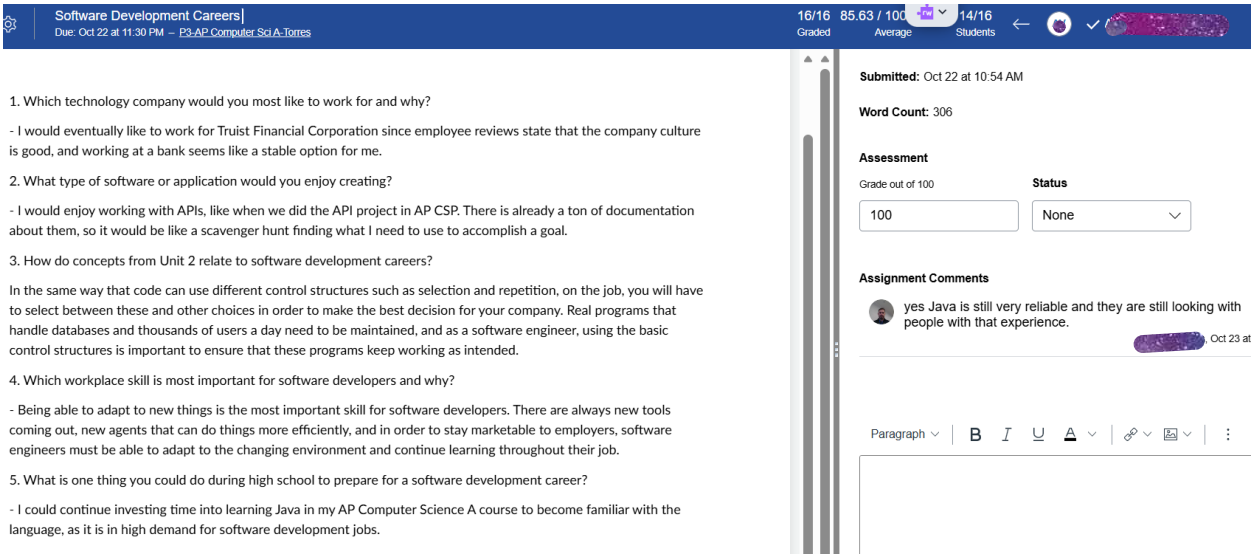
FBLA/Career Development Connection: Students analyzed industry careers and reflected on communication, teamwork, professionalism, leadership, and problem-solving skills required for success in game development and FBLA competitive events.

Student Sample Evidence and teacher Feedback



AP Computer Science A

Activity Summary: Software Development Careers activity. Students researched Java-related careers in Metro Atlanta, examined job requirements and workplace expectations, and connected Unit 2 programming concepts to professional software development roles.
FBLA/Career Development Connection: Students explored software development career pathways while identifying workplace communication, collaboration, professionalism, and problem-solving skills aligned with FBLA and industry expectations.

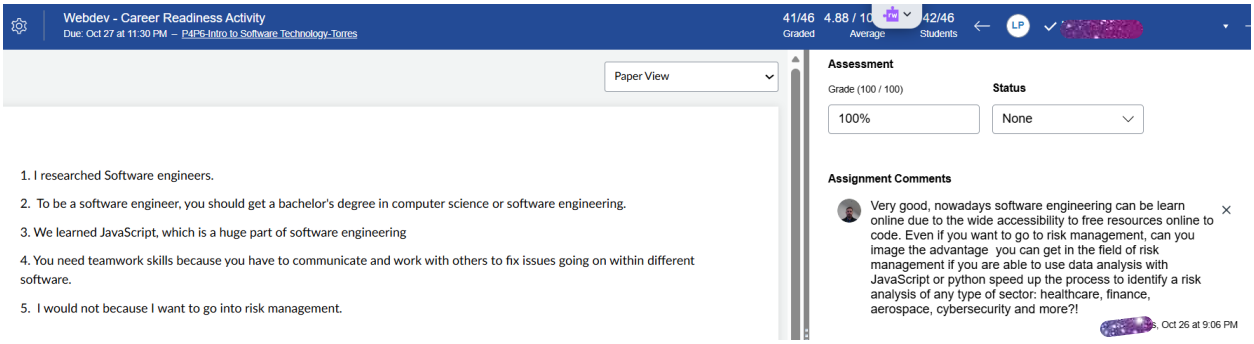


Introduction to Software Technology / CSP

Activity Summary: Career Exploration Website Reflection activity. Students reflected on technology careers researched during their website project and evaluated required education, certifications, technical skills, and career opportunities.

FBLA/Career Development Connection: Students investigated technology career pathways and demonstrated understanding of leadership, communication, teamwork, professionalism, and career planning skills emphasized in FBLA.

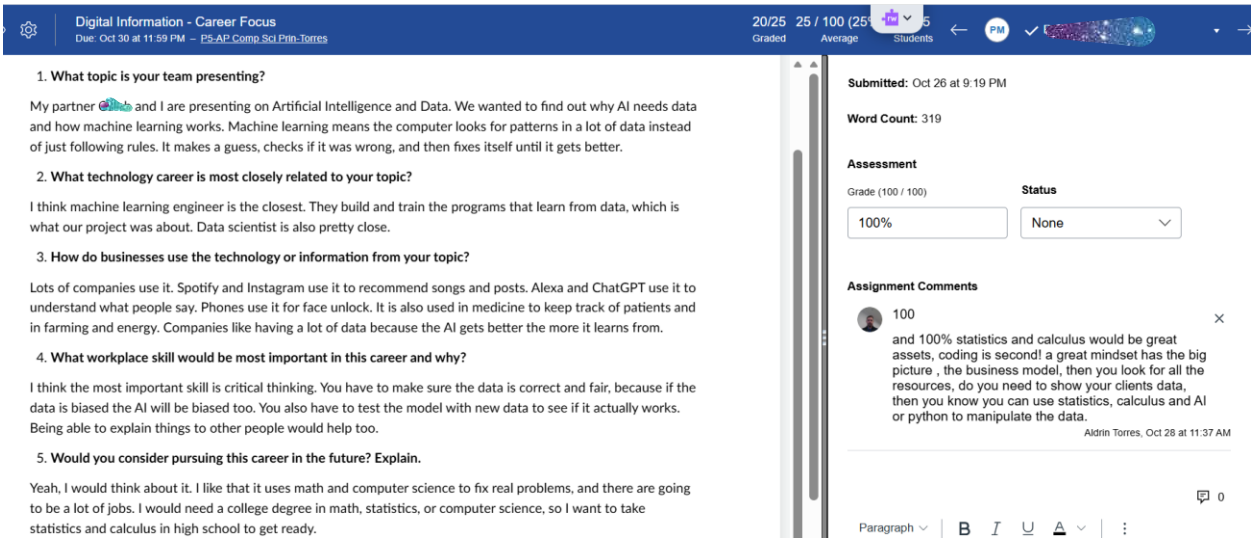
Student Sample Evidence and Teacher Feedback:



AP Computer Science Principles

Activity Summary: Technology Careers in Metro Atlanta activity. Students researched careers in data science, artificial intelligence, cybersecurity, and software development while connecting course concepts to workforce opportunities.

FBLA/Career Development Connection: Students explored high-demand technology careers and identified workplace communication, critical thinking, professionalism, and career readiness skills aligned with FBLA competitive events and industry standards.


Digital Information - Career Focus
Due: Oct 30 at 11:59 PM – P5-AP Comp Sci Prin-Torres
20/25
Graded
25 / 100 (25%
Average
5
Students
PM
1. What topic is your team presenting?
My partner and I are presenting on Artificial Intelligence and Data. We wanted to find out why AI needs data and how machine learning works. Machine learning means the computer looks for patterns in a lot of data instead of just following rules. It makes a guess, checks if it was wrong, and then fixes itself until it gets better.
2. What technology career is most closely related to your topic?
I think machine learning engineer is the closest. They build and train the programs that learn from data, which is what our project was about. Data scientist is also pretty close.
3. How do businesses use the technology or information from your topic?
Lots of companies use it. Spotify and Instagram use it to recommend songs and posts. Alexa and ChatGPT use it to understand what people say. Phones use it for face unlock. It is also used in medicine to keep track of patients and in farming and energy. Companies like having a lot of data because the AI gets better the more it learns from.
4. What workplace skill would be most important in this career and why?
I think the most important skill is critical thinking. You have to make sure the data is correct and fair, because if the data is biased the AI will be biased too. You also have to test the model with new data to see if it actually works. Being able to explain things to other people would help too.
5. Would you consider pursuing this career in the future? Explain.
Yeah, I would think about it. I like that it uses math and computer science to fix real problems, and there are going to be a lot of jobs. I would need a college degree in math, statistics, or computer science, so I want to take statistics and calculus in high school to get ready.
Submitted: Oct 26 at 9:19 PM
Word Count: 319
Assessment
Grade (100 / 100)
Status
100%
None
Assignment Comments
100
and 100% statistics and calculus would be great assets, coding is second! a great mindset has the big picture , the business model, then you look for all the resources, do you need to show your clients data, then you know you can use statistics, calculus and AI or python to manipulate the data.
Aldrin Torres, Oct 28 at 11:37 AM
0
Paragraph